Electrical load list + hazardous-area table

SO3 Converter Ver 1 · SO3-V1-EE-LL-001 · Rev 0

Department of Chemical Engineering, Chulalongkorn University (teaching)

REV	DATE	DESCRIPTION	ORIGINATOR	CHECKED / APPROVED
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Contents

1 · Purpose and scope 3

2 · Basis 3

3 · Electrical load list 4

4 · Single-line diagram, transformers, UPS and K-101 starting 5

5 · Cable sizing (main feeders) 6

6 · Hazardous-area classification 6

7 · Checks against the model and the other documents 7

8 · Engineering queries 7

1 · Purpose and scope

This document lists the electrical loads of the SO3 Converter Ver 1 converter section, sketches the supply (single-line diagram), sizes the transformers, the UPS for the control and safety systems and the main cables, and classifies the hazardous areas. K-101 power is the simulator model value at the design point (econ(), the Efficiency tab); pump powers come from the Mechanical datasheets SO3-V1-ME-EQL-001. Supply data, efficiencies, power factors, cable data and the H_2S basis are assumed or typical.

2 · Basis

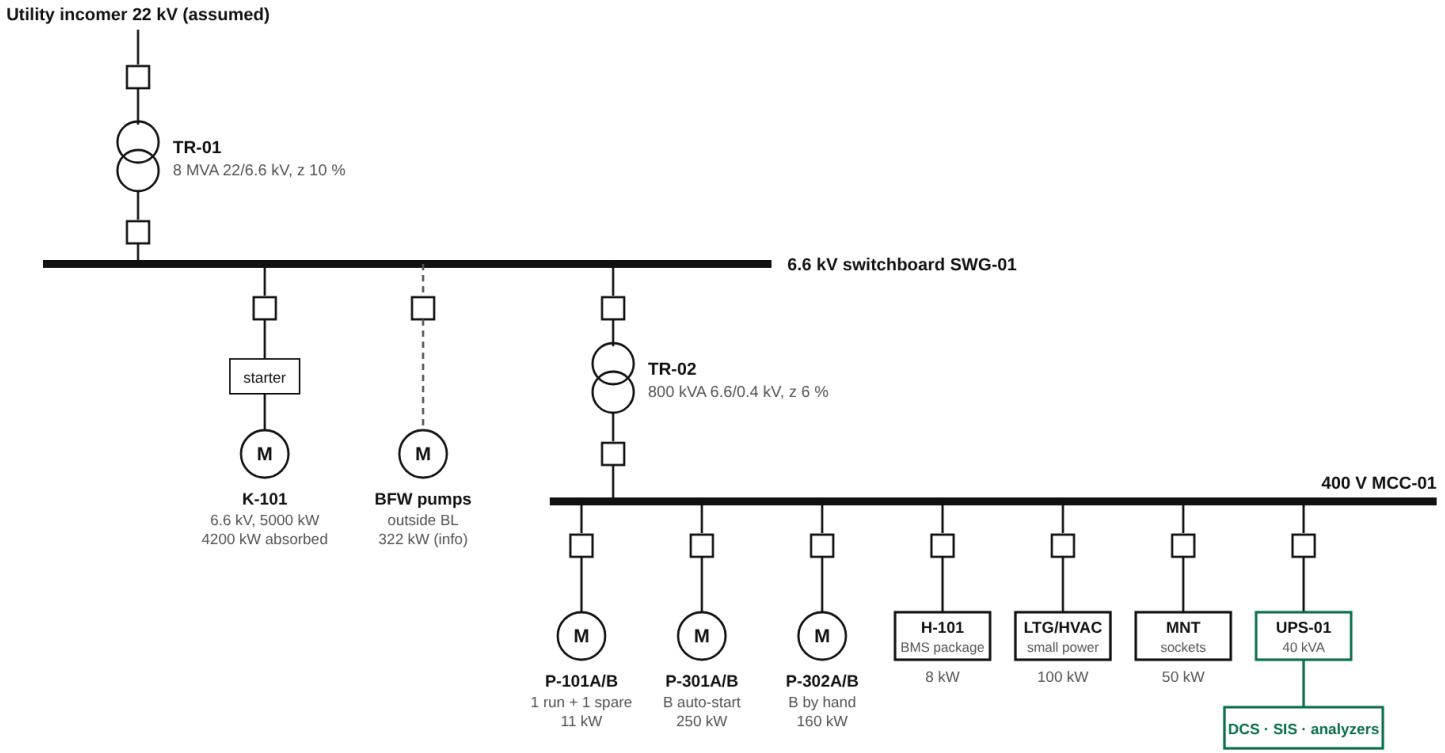
- Electrical input = absorbed kW / motor efficiency; kVA = input / pf.
- Consumed load = input × load factor: continuous 1.0, intermittent 0.3, standby 0 (typical practice; spares never added to the running load).
- Motor rating ≥ 1.1 × absorbed power, next standard size (typical). Motors ≥ 300 kW at 6.6 kV, smaller at 400 V; DCS/SIS/analyzers at 230 V from UPS-01.
- Transformers loaded to ≤ 80 % of rating on the consumed load; UPS rated ≥ 1.25 × its load with 30 min battery (typical).
- Utility incomer 22 kV, upstream fault level 500 MVA, TR-01 z 10 %, TR-02 z 6 % (assumed).
- Hazardous areas: IEC 60079-10-1 (gas), IEC 60079-10-2 (dust); NFPA 497/499 and API RP 505 as equivalents (typical).

3 · Electrical load list

Tag	Service	Supply	Duty	Spare start	Abs. kW	Rating kW	η	pf	Input kW	kVA	FLC A	LF	Cons. kW	Cons. kVA	Area
K-101	Main air blower motor	6.6 kV MV	cont.	—	4200.0	5000	0.97	0.88	4329.9	4920.3	512	1.0	4329.9	4920.3	A5
P-101A	Molten sulfur pump (running)	400 V LV	cont.	—	8.5	11	0.88	0.82	9.7	11.8	22	1.0	9.7	11.8	A2
P-101B	Molten sulfur pump (spare, started by hand)	400 V LV	standby	manual	8.5	11	0.88	0.82	9.7	11.8	22	0.0	0.0	0.0	A2
P-301A	IPA acid pump (running)	400 V LV	cont.	—	225.0	250	0.95	0.87	236.8	272.2	437	1.0	236.8	272.2	A6
P-301B	IPA acid pump (spare, auto-start HS-301B)	400 V LV	standby	auto	225.0	250	0.95	0.87	236.8	272.2	437	0.0	0.0	0.0	A6
P-302A	FAT acid pump (running)	400 V LV	cont.	—	141.0	160	0.95	0.86	148.4	172.6	283	1.0	148.4	172.6	A6
P-302B	FAT acid pump (spare, started by hand)	400 V LV	standby	manual	141.0	160	0.95	0.86	148.4	172.6	283	0.0	0.0	0.0	A6
H-101	Preheat burner package: igniter, scanners and cooling fans, BMS panel, fuel valves (start-up only; combustion air from K-101)	400 V LV	interm.	—	8.0	8	0.90	0.85	8.9	10.5	15	0.3	2.7	3.1	A4
DCS	DCS controllers, I/O, operator stations	230 V UPS	cont.	—	12.0	12	1.00	0.90	12.0	13.3	—	1.0	12.0	13.3	—
SIS	SIS logic solver and I/O (I-1…I-6), XV-103 solenoids	230 V UPS	cont.	—	4.0	4	1.00	0.90	4.0	4.4	—	1.0	4.0	4.4	—
ANZ	Analyzers AI-104, AI-105, AIT-306/307, AI-303 and shelters	230 V UPS	cont.	—	8.0	8	1.00	0.90	8.0	8.9	—	1.0	8.0	8.9	—
F&G/TEL	SO_2 gas detection, field instruments, telecom	230 V UPS	cont.	—	3.0	3	1.00	0.90	3.0	3.3	—	1.0	3.0	3.3	—
LTG	Lighting and small power	400 V LV	cont.	—	40.0	40	1.00	0.90	40.0	44.4	—	1.0	40.0	44.4	—
HVAC	Substation and control-room HVAC (pressurised, filtered intake)	400 V LV	cont.	—	60.0	60	1.00	0.85	60.0	70.6	—	1.0	60.0	70.6	—
MNT	Maintenance and welding sockets	400 V LV	interm.	—	50.0	50	1.00	0.80	50.0	62.5	—	0.3	15.0	18.8	—
BFW-A/B	Boiler feed-water pumps (steam system, outside battery limit; listed for information)	6.6 kV MV	cont.	—	322.0	355	0.95	0.87	338.9	389.6	38	1.0	(338.9) not in total	—	—
Total	in the battery limit								5306	6052			4870	5544	

Sources: K-101 = model econ() / Efficiency tab; P-101 = model sulfur supply pressure × flow / 0.45; P-301/P-302 = Mechanical datasheets (22 m head, η 0.75); other loads assumed.

4 · Single-line diagram, transformers, UPS and K-101 starting



Notes: breakers shown as squares; drives as M. Ratings from the load list; transformer and incomer data assumed.
Dashed: outside the battery limit (not in the total). UPS: double conversion, 30 min battery (typical).

SINGLE-LINE DIAGRAM (SKETCH) · SO3-V1-EE-LL-001 · Rev 0

Item	Load	Selected
TR-01 22/6.6 kV	6930 kVA (consumed / 0.8)	8000 kVA, z 10 % (assumed)
TR-02 6.6/0.4 kV	779 kVA	800 kVA, z 6 % (assumed)
UPS-01	30.0 kVA × 1.25	40 kVA, 30 min
K-101 start, direct on line	35145 kVA	6.6 kV bus dip 33.8 %
K-101 start, soft starter	17572 kVA	dip 20.3 %
K-101 start, VSD	6443 kVA	dip 8.5 %

5 · Cable sizing (main feeders)

I = P / (√3 V pf η); running drop ΔV% = √3 I L (R cosφ + X sinφ) / V × 100 ≤ 5 % (motors) / 3 % (feeders); starting drop at 6 × I, pf 0.3 ≤ 15 %; derating 0.8 (assumed). Copper XLPE, typical teaching data (check a catalogue). The tab has the interactive calculator.

Feeder	kW	kV	L m	FLC A	Runs	Selected	ΔV run %	ΔV start %
K-101	5000	6.6	150	512	2	2 × 3c 120 mm² MV	0.22	0.9
P-301A	250	0.4	120	437	2	2 × 3c 95 mm² LV	2.88	10.2
P-302A	160	0.4	120	283	1	1 × 3c 150 mm² LV	2.61	10.9
P-101A	11	0.4	150	22	1	1 × 3c 6 mm² LV	4.68	10.8
TR-02 LV side	720	0.4	30	1155	6	6 × 3c 70 mm² LV	0.86	—

6 · Hazardous-area classification

The converter train carries no flammable gas (SO_2, SO_3, O_2, N_2: toxic or oxidising); acid areas are corrosive, not hazardous. Flammable sources: H_2S from molten sulfur (assumed, not in the model), sulfur dust, start-up fuel gas.

Area	Location	Source	Grade	Zone	Group	T-class	Extent	Reasoning	Ex rating
A1	Molten sulfur pit TK-101: vapour space inside (outside battery limit)	H_2S released from molten sulfur (assumed: sulfur not degassed), sulfur vapour; sulfur sublimate on the roof	continuous / primary	Zone 1 (gas) + Zone 21 (dust)	IIB · IIIB	T3 · T135 °C	whole vapour space; pit sweep air keeps H_2S below 25 % LFL (assumed)	H_2S LFL 4 %, AIT 260 °C → IIB T3 (T3 = 200 °C max surface); sulfur dust cloud ignition ≈ 190 °C → surface ≤ 2/3 of it	Ex db IIB T3 / Ex tb IIIB T135 °C
A2	Pit roof, P-101A/B motors, pit vents and openings, 3 m radius	H_2S and sulfur vapour from vents; spilled and solidified sulfur	secondary	Zone 2 + Zone 22	IIB · IIIB	T3 · T135 °C	3 m from vents and openings, 1 m from pump glands (typical)	release only in abnormal operation (vent upset, gland leak); good natural ventilation	Ex ec IIB T3 / Ex tc IIIB T135 °C
A3	Sulfur filters FL-101A/B and jacketed sulfur lines	filter opening for cleaning (cake), flanged joints	secondary	Zone 2 + Zone 22 within 1 m of flanges / filter covers; else non-hazardous	IIB · IIIB	T3 · T135 °C	1 m (typical)	welded steam-jacketed lines; release only at joints and when a filter is opened	Ex ec IIB T3 / Ex tc IIIB T135 °C
A4	F-101 burner front: sulfur guns, H-101 burner and its fuel-gas train	H_2S at sulfur-gun connections; natural gas (assumed fuel) at the fuel-train flanges and vents	secondary	Zone 2	IIB (covers IIA natural gas)	T3	1 m around guns and fuel-train flanges, 3 m around vent tips (typical)	fuel gas is the only flammable gas of the unit and is used at start-up only; H_2S sets the group	Ex ec IIB T3
A5	K-101 blower house; F-101 shell, B-101, R-201, SH-201, E-201…E-205, gas ducts	process gas $SO_2/SO_3/O_2/N_2$ (toxic, oxidising, not flammable)	—	Non-hazardous	—	—	—	no flammable gas in the converter train: a leak is a toxic release (SO_2 detection), not an ignition hazard	none
A6	T-301, T-302, TK-301/302, P-301A/B, P-302A/B, acid coolers	98.5 % sulfuric acid (corrosive)	—	Non-hazardous (corrosive)	—	—	—	acid is not flammable. Hydrogen can form where diluted acid meets carbon steel (washing, water ingress): handled by gas test and permit before entry or hot work, not by area classification (typical)	none
A7	Substation, control room, UPS and battery room	VRLA batteries (hydrogen on charge)	—	Non-hazardous (ventilated)	—	—	—	battery room ventilation to IEC 62485-2 keeps H_2 far below LFL; SO_2 detection at the HVAC intake	none
A8	Solid sulfur receipt and melter (outside battery limit)	sulfur dust	primary / secondary	Zone 21 / 22 (by others)	IIIB	T135 °C	by others	IEC 60079-10-2 dust classification	by others

7 · Checks against the model and the other documents

14 of 16 checks pass. The same checks are recomputed live in the Electrical tab of the simulator.

Check	Expected	Found	Result	Source
K-101 absorbed power in the load list = model blower power (econ at design)	4200 kW	4200 kW	✓	econ(copySim(snapDesign))
K-101 = Efficiency tab blower power at design	4200 kW	4200 kW	✓	EF_D.des.kW
K-101 motor rating ≥ hand-check shaft power at η 0.80 (EQ-ee-01)	≥ 4978 kW	5000 kW	✓	Mechanical datasheet
Total consumed load = Σ rows × load factor	4869 kW	4869 kW	✓	load list
Standby drives add nothing to the consumed load	0 kW from 3 standby drives	0 kW	✓	EE_LF.S = 0
Running / standby in the list = pumps running in the model at design	P-101A run/B off; P-301A run/B off; P-302A run/B off	same	✓	snapDesign P.sw
Spare start mode = model (P-301B auto; P-302B, P-101B by hand)	P-301B auto, P-302B manual, P-101B manual	P-301B auto, P-302B manual, P-101B manual	✓	snapDesign P.auto
Spare start mode = Reliability tab series model	P-301B auto, P-302B manual, P-101B by hand	same (running; P-302B manual only)	✓	RL_EL (Process reliability)
SIS and DCS are fed from the UPS	SIS and DCS on UPS	SIS ✓ DCS ✓	✓	HAZOP R36
UPS rating ≥ 1.25 × UPS load	≥ 37.5 kVA	40 kVA, 30 min	✓	typical sizing margin
SIS de-energises to trip (safe state on loss of power)	de-energise to trip	yes (SRS common requirements)	✓	SRS sf.json
LV transformer ≥ LV + UPS consumed load / 0.8 loading	≥ 779 kVA	800 kVA	✓	load list
Main transformer ≥ total consumed kVA / 0.8	≥ 6930 kVA	8000 kVA	✓	load list
K-101 direct-on-line start: MV bus dip ≤ 15 % (typical)	≤ 15 %	33.8 % (DOL, 6 × FLC)	✗	EQ-ee-03
LV motors within the LV practice limit (FLC ≤ 300 A at 400 V, typical)	≤ 300 A	P-301A 437 A	✗	EQ-ee-02
Every drive has an area classification and an Ex rating that fits it	8 drives	all fit	✓	part 5

8 · Engineering queries

No.	Document	Subject	Finding	Proposed
EQ-ee-01	SO3-V1-EE-LL-001	K-101 motor rating basis	The load list uses the model blower power 4200 kW (absorbed). The Mechanical hand check (EQ-em-04) implies η 0.95; at η 0.80 the shaft power would be 4978 kW. The 5000 kW motor covers it with 0.5 % margin only.	Hold the motor rating at 5000 kW until the vendor curve is in; confirm before ordering the motor and the 6.6 kV feeder.
EQ-ee-02	SO3-V1-EE-LL-001	P-301A/B motor voltage	P-301A/B 250 kW at 400 V draws 437 A full load (≈ 2.6 kA starting), above the 300 A LV practice limit used here.	Supply P-301A/B at 690 V (≈ 253 A) or 6.6 kV, or keep 400 V with a soft starter and parallel cables; decide with the MCC design.
EQ-ee-03	SO3-V1-EE-LL-001	K-101 starting	Direct-on-line start of the 5000 kW K-101 (6 × FLC, 35.1 MVA) on a 8 MVA, 10 % transformer dips the 6.6 kV bus by about 33.8 % (typical limit 15 %). Soft start (≈ 3 × FLC) 20.3 %; VSD 8.5 %.	Use a VSD on K-101 (also the energy option in the Efficiency tab) or a soft starter; check the upstream fault level (assumed 500 MVA at 22 kV).
EQ-ee-04	SO3-V1-EE-LL-001	H_2S basis of the hazardous-area classification	The model has no H_2S. The sulfur-pit classification (Zone 1 inside, Zone 2 around vents, IIB T3) assumes undegassed molten sulfur with dissolved H_2S and a pit sweep-air system that is not on the P&IDs.	Confirm the sulfur specification (degassed or not) and the pit ventilation design; revise the zones and the Ex ratings of P-101A/B accordingly.